

Assignment 2

Guidelines:

1. Logistics. Your solution must be submitted as a *typed* PDF document. Scanned handwritten solutions, solutions in any other format, or unreadable solutions will not be accepted. You are encouraged to use L^AT_EX.

Please submit a separate PDF for each question.

Submit your solution in Crowdmark by **Friday, Apr 4, 23:59**. If you email the instructor before the deadline to ask for an extension, an extension may be given until **Tuesday, Apr 8, 23:59**. You are strongly encouraged to submit the solution *several days in advance* – last-minute technical problems in your systems and/or in Crowdmark will not grant any extension.

2. Answering questions. In all assignments and exams in this course, unless specifically and explicitly stated otherwise:

- Establish correctness for all of your answers using **rigorous proofs**. Your solution will be marked based on its completeness and correctness as well as on the correctness, clarity and precision of your proofs.
- Describe algorithms in high-level pseudocode. Do not provide low-level executable code. On the other hand, if your code is overly abstract and does not sufficiently specify relevant parts (e.g., it is not obvious how to efficiently implement parts), you may lose marks.
- When you are asked to prove an upper bound on the running time of an algorithm, the intention is to prove the tightest upper bound that you can. Non-tight upper bounds may receive partial (or no) marks.

3. Academic integrity. Assignments are to be solved individually, and according to the course integrity policy and the Code of Behaviour on Academic Matters. Here is a reminder of one of the relevant parts from the syllabus:

Large assignments should be solved individually, and written by yourself, without imitating online texts and without making use of generative AI (or any other software or automated tool). Any submitted assignment that is suspiciously similar to another assignment or to online texts will be automatically investigated as an integrity violation.

You are encouraged to ask questions on Piazza, without directly asking how to solve a problem or revealing the solution. You may not consult any other resources except your class notes, your textbook, Piazza, and assigned readings. Consulting any other resource or collaborating with other students is a violation of the academic integrity policy.

Question 1. We are given a directed weighted graph $G = (V, E)$, where each edge has weight either 1 or 2, and a source node $s \in V$. Our goal is to find the shortest paths from s to any other node in the graph.

1. What is the fastest algorithm we learned for this problem? What is its runtime?
2. Show a faster algorithm (relying on the fact that we're dealing with the special case in which weights are either 1 or 2).

(10pts.)

Question 2. Design an algorithm that solves the All Pairs Shortest Paths problem using dynamic programming. Your algorithm should run in time $O(|V|^3)$. Compare the runtime of your algorithm to the runtime of Johnson's algorithm: On which type of graphs does one algorithm outperform the other? (10pts.)

Question 3. For a given flow network (G, c, s, t) , we say that a flow f is integer-valued if $f(e) \in \mathbb{N} \cup \{0\}$ for every $e \in E$. Prove that for every flow network¹ there exists a maximum flow that is integer-valued (i.e., the flow achieves the maximum possible flow value, and is also integer-valued). Also prove that there exists a flow network with a maximum flow that is not integer-valued. (5pts.)

Question 4. We are given a flow network where all edges have capacity exactly 1, and an integer $k \in \mathbb{N}$ that is at most the maximum flow in the network. Design a fast algorithm that deletes k edges from the graph to obtain another network in which the maximum flow is smaller by exactly k . (20pts.)

Question 5. Consider the following problem.

We are given a matrix $M \in \mathbb{R}^{n \times m}$. We want to find a vector $x \in \mathbb{R}^n$ with minimum ℓ_2 norm (i.e., minimum value of $\|x\|_2 = \sqrt{\sum_{i \in [n]} x_i^2}$) such that for every column $i \in [m]$ in M it holds that $\sum_{j \in [n]} M_{j,i} \cdot x_j^2 \geq -1$.

Show how to solve the problem above using linear programming:

1. Write a linear program that allows to solve the problem. Prove that the optimal value for this linear program can be converted (by a single arithmetic operation) to the minimum ℓ_2 norm of a solution for the problem above.
2. Convert your linear program to standard form, and explicitly write the resulting matrix and vectors (A, b, c) . Explain why the standard form is equivalent to the original LP.

(15pts.)

¹Recall that we only allowed integer capacities in our flow networks.

Question 6. In the tutorials you saw an LP relaxation for the Vertex Cover problem.

1. Show an explicit graph $G = (V, E)$ such that the value of the optimal solution for the LP relaxation is different than the minimum size of a vertex cover for G .

Definition. We define the Integrality Gap of a minimization-based LP (A, b, c) over $\mathbb{R}^n$ as $\frac{M^{\text{int}}}{M^{\text{frac}}}$, where

$$M^{\text{int}} = \min_{x \in \{0,1\}^n, A \cdot x \leq b} \left\{ c^T \cdot x \right\} ,$$

$$M^{\text{frac}} = \min_{x \in \mathbb{R}^n, A \cdot x \leq b, x \geq 0^n} \left\{ c^T \cdot x \right\} .$$

2. For every $n \in \mathbb{N}$, show a graph with n vertices such that the integrality gap of the vertex cover LP relaxation is at least $2 \cdot (1 - 1/n)$.
3. Let $x \in \mathbb{R}^n$ be a feasible vector for the vertex cover LP relaxation of an given n -vertex graph. Prove that we can round x to a *feasible* vector $y \in \{0,1\}^n$ such that $c^T \cdot y \leq 2c^T \cdot x$.
4. What bounds do Items #2 and #3 give for the integrality gap of the vertex cover LP relaxation?
5. If you want to get an approximation for the minimum vertex cover size, would you use the LP+rounding algorithm from Item #3, or the algorithm you analyzed in Assignment 1? Why?

(25pts.)

Question 7. Given a bipartite graph $G = (V, E)$, we define:

1. A fractional matching is an assignment of non-negative real numbers to edges $m: E \rightarrow \mathbb{R}^+$ such that for each $v \in V$ we have $\sum_{e \ni v} m(e) \leq 1$. The value of a fractional matching is $\sum_{e \in E} m(e)$.
2. A fractional vertex cover is an assignment of non-negative real numbers to vertices $c: V \rightarrow \mathbb{R}^+$ such that for each $e = (u, v) \in E$ we have $c(u) + c(v) \geq 1$. The value of a fractional vertex cover is $\sum_{v \in V} c(v)$.

Prove that for every bipartite graph, the maximum value of any fractional matching equals the minimum value of any fractional vertex cover. (15pts.)